## Description

### Illustration





### Figure caption

Engineering brick is used in applications where high strength or low water absorption are critical. 1. The Welland Valley viaduct, England. 2. A brick-lined underground sewer. Images used under license from Shutterstock.com

### The material

Brick is as old as Babylon (4000 BC) and as durable. Palaces and temples of Mesopotamia were built of stone and sun-dried brick. It is the most ancient of all man-made building materials and belongs to that equally ancient building method of masonry construction. The regularity and proportions of bricks makes them easy to lay in a variety of patterns, and their durability and relative low cost makes them an ideal material for the construction of buildings. Clay - the raw material from which bricks are made - is available almost everywhere; finding the energy to fire them can be more of a problem. Pure clay is gray-white in color; the red color of most bricks comes from impurities of iron oxide. Yellow, gray and even black bricks can be made by varying the proportion of various clays and other ingredients and carefully controlling the heat of firing.

Brick is used mainly for walls and foundations. Common brick is used where appearance is not important, or where it will be plastered or rendered. Engineering brick is used where high strength, low porosity and high frost resistance are required. Facing brick is used for external work where good appearance is needed. Brick walls may be structural or non-structural; the latter being much more common as a veneer material spanning between the columns of a structural frame.

The quoted properties are those for a single brick rather than that of brickwork, in which the properties and adhesion of mortar would have to be taken into consideration (most obviously becoming a dominant factor when loading in tension).

### Compositional summary

50-80% alumina and 20-35% silica

## General properties

| | | | | |
|---|---|---|---|---|
| Density | 2.1e3 | - | 2.5e3 | kg/m^3 |
| Price | * 1.62 | - | 2.88 | GBP/kg |

## Material form that data applies to

| | | | | | |
|---|---|---|---|---|---|
| Bulk | ✓ | | | | |

## Building system

| | | | | | |
|---|---|---|---|---|---|
| Superstructure | ✓ | | | | |
| Enclosure | ✓ | | | | |
| Interiors | ✓ | | | | |
| Services | ✗ | | | | |

## Mechanical properties

| | | | | | |
|---|---|---|---|---|---|
| Young's modulus | | 30 | - | 35 | GPa |
| Shear modulus | * | 12.5 | - | 14.6 | GPa |
| Bending modulus | * | 30 | - | 35 | GPa |
| Poisson's ratio | * | 0.2 | - | 0.23 | |
| Yield strength (elastic limit) | | 5 | - | 15 | MPa |
| Tensile strength | | 5 | - | 15 | MPa |
| Compressive strength | | 45 | - | 150 | MPa |
| Bending strength | | 5 | - | 15 | MPa |
| Elongation | * | 0.01 | - | 0.05 | % strain |
| Hardness - Vickers | | 15 | - | 45 | HV |
| Fatigue strength at 10^7 cycles | * | 2 | - | 6 | MPa |
| Fracture toughness | | 0.00143 | - | 0.015 | MPa.m^0.5 |
| Mechanical loss coefficient (tan delta) | * | 0.00325 | - | 0.0036 | |

## Thermal and combustion properties

| | | | | | |
|---|---|---|---|---|---|
| Thermal conductor or insulator? | | Poor insulator | | | |
| Thermal resistivity | | 1 | - | 1.25 | m.°C/W |
| Thermal expansion coefficient | | 5 | - | 8 | µstrain/°C |
| Specific heat capacity | | 750 | - | 850 | J/kg.°C |
| Melting point | | 1.2e3 | - | 1.5e3 | °C |
| Maximum service temperature | | 977 | - | 1.18e3 | °C |
| Flammability | | Non-flammable | | | |
| Emissivity | | 0.75 | - | 0.93 | |

## Hygro-thermal properties

| | | | | | |
|---|---|---|---|---|---|
| Water absorption | * | 0.5 | - | 2 | % |
| Water vapor permeability | | 4.38e-12 | - | 4.67e-12 | kg.m/s.m^2.Pa |
| Frost resistance | | Good | | | |

## Electrical properties

| | | | | |
|---|---|---|---|---|
| Electrical conductor or insulator? | Good insulator | | | |
| Electrical resistivity | 1e14 | - | 3e14 | µohm.cm |
| Dielectric constant (relative permittivity) | 4 | - | 10 | |
| Dissipation factor (dielectric loss tangent) | 0.001 | - | 0.01 | |

## Optical properties

### Natural color

Red or gray

| | | | | |
|---|---|---|---|---|
| Transparency | Opaque | | | |
| Transmissivity | 0 | | | % |

## Critical Materials Risk

| | |
|---|---|
| High critical material risk? | Yes |

## Acoustic properties

| | |
|---|---|
| Sound absorption | Poor |
| Sound isolation | Good |

## Processability (scale 1 = impractical to 5 = excellent)

| | | | |
|---|---|---|---|
| Moldability | 2 | - | 4 |
| Machinability | 1 | - | 2 |

## Durability

| |
|---|
| Water (fresh) | Excellent |
| Water (salt) | Excellent |
| Weak acids | Acceptable |
| Strong acids | Unacceptable |
| Weak alkalis | Acceptable |
| Strong alkalis | Unacceptable |
| Organic solvents | Excellent |
| UV radiation (sunlight) | Excellent |
| Wear resistance | Limited use |
| Industrial atmosphere | Acceptable |
| Rural atmosphere | Excellent |
| Marine atmosphere | Excellent |

## Primary material production: energy and climate change

| | | | | | |
|---|---|---|---|---|---|
| Climate change ($CO_2$-eq) (virgin grade production) | * | 0.188 | - | 0.208 | kg/kg |
| Embodied energy (virgin grade production) | * | 3.27 | - | 3.61 | MJ/kg |

| | | | | |
|---|---|---|---|---|
| Water Usage | * 5.27 | - | 5.83 | l/kg |

## Material processing: energy

| | | | | |
|---|---|---|---|---|
| Machining, grinding, embodied energy | * 20.9 | - | 35.4 | MJ/kg |

## Material processing: climate change

| | | | | |
|---|---|---|---|---|
| Machining, grinding, climate change (CO2-eq) | * 1.36 | - | 2.32 | kg/kg |

## Recycling and end of life

| | | | | |
|---|---|---|---|---|
| Recycle | ✓ | | | |
| Functional recycle | ✓ | | | |
| Climate change (CO2-eq), recycling | * 0.0188 | - | 0.0208 | kg/kg |
| Embodied energy, recycling | * 0.458 | - | 0.505 | MJ/kg |
| Recycle fraction in current supply | 0 | | | % |
| Combust for energy recovery | ✗ | | | |
| Landfill | ✓ | | | |
| Biodegrade | ✗ | | | |
| A renewable resource? | ✗ | | | |

## Supporting information

### Design guidelines

Bricks and the mortar used to bond them are strong in compression but weak in tension. This largely determines the way they are used: brick structures (like those of stone) are designed in such a way that the loads are compressive everywhere, ruling out cantilevered or very slender forms. The face of brick can be molded and glazed, allowing great freedom of decoration, and the pattern and natural color variation gives a visually attractive surface. Low-fired, unglazed brick is vulnerable to water-penetration and degradation, and should be screened from direct weathering. High-fired or glazed brick is extremely durable. The durability of brick walls, as a building system, is dependent on the proper composition and tooling of the mortar joint and adequate weep holes that allow trapped liquid water to escape to the outside from inside the brick cavity. Freeze-thaw cycles, in which trapped water expands, are a common mechanism for cracked brick and spalled mortar.

The standard brick size is 3 1/2 in. (w) x 2 1/4" in. (d) x 8" in. (h), with relatively large tolerances. Modular bricks are sized such that three courses plus mortar joints amount to 8 inches in height. Modular, engineer modular, engineer standard, Roman, Norman, King size and other standard sizes are used for a variety of purposes and reasons. Non-standard sizes are also available. Solid, cored, hollow and 'frogged' bricks are all widely available.

Three grades of building and facing bricks are available in the US; Grade SW for use in conditions of severe weathering, MW for moderate weathering and NW for negligible weathering. Three types of facing brick are used; Type FBX (highly consistent dimensional and color uniformity), Type FBS (wide color variation and less dimensional consistency), and Type FBA (great color and dimensional variations).

### Technical notes

Brick is fired clay: hydrous aluminum silicate, with impurities of potash, soda, lime, and oxides of iron. On heating to 900-1200C, the aluminum silicate reacts with soda to form a viscous glass - it is this process that bonds the brick together. Most brick is porous; if it is to resist water it is glazed. To do this the surface of the fired brick is painted with a mixture of glass-forming fluxes and the brick is re-fired, melting the glaze, and forming a glassy surface that can be colored.
Large pores and low porosity are the most important factor in establishing if a particular brick type will be durable.
The quoted properties are those for a single brick rather than that of brickwork, in which the properties and adhesion of mortar would have to be taken into consideration (most obviously becoming a dominant factor when loading in tension).

## Typical uses

Engineering construction, external paving, capping bricks for walls.

## Related standards

mortar: ASTM C144 (mortar sand), ASTM C150 (Portland cement), ASTM C207 (lime), ASTM C91 (masonry cements), ASTM C270 (mortar composition)
brick: ASTM C62, C216, C652 (grades and types of brick), ASTM C902 (paving bricks)
BS 3921 (1969)
CSI/CSC Masterformat Section Numbers:
04 05 16 Masonry Grouting
04 05 23 Masonry Accessories
04 20 00 UNIT MASONRY
04 21 00 Clay Unit Masonry
04 23 00 Glass Unit Masonry
04 50 00 REFRACTORY MASONRY
04 51 00 Flue Liner Masonry
04 57 00 Masonry Fireplaces

## Links

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